

FIELD THEORY PART-II

Phase 3: Divergence Mastery

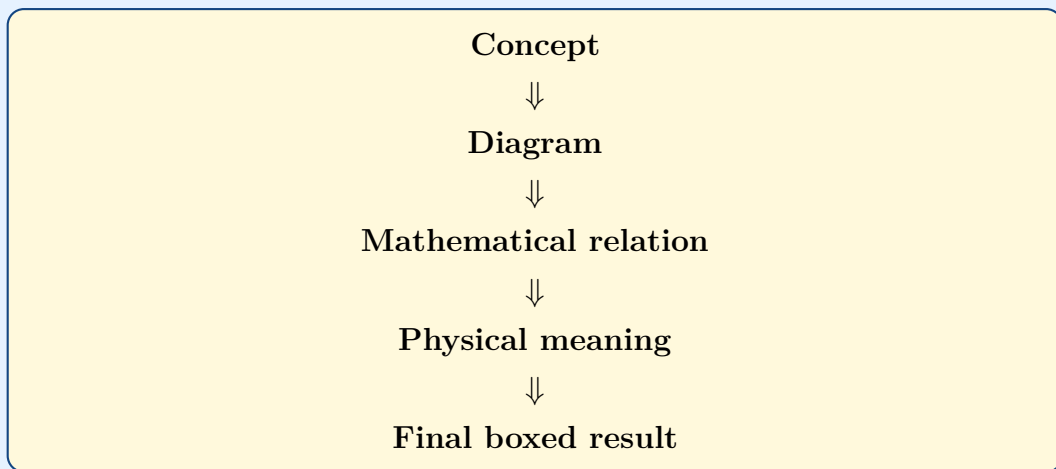
Source, sink, divergence of B and divergence of J

Academic Article Style Notes for Semester Preparation

How to Use This Phase

This phase is made for semester preparation from a very basic level. Read the concept first, then understand the diagram, then learn the derivation pattern, and finally practise the exam-style writing block.

The recommended writing order is:



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1 Aim of Phase 3

Main Goal

Divergence is needed to understand $\nabla \cdot \mathbf{B} = 0$, $\nabla \cdot \mathbf{J} = 0$, conservation of charge, and the idea of source or sink of a field.

2 Definition of Divergence

Fundamental Definition

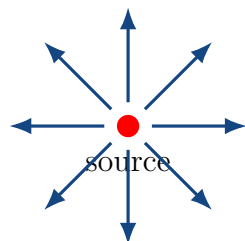
Divergence of a vector field $\mathbf{A}$ is the net outward flux per unit volume:

$$\nabla \cdot \mathbf{A} = \lim_{\Delta V \rightarrow 0} \frac{1}{\Delta V} \oint_S \mathbf{A} \cdot d\mathbf{S}$$

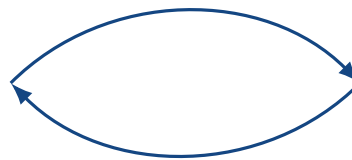
3 Physical Meaning

Source and Sink Meaning

If more field comes out than enters, divergence is positive. If more field enters than comes out, divergence is negative. If field entering equals field leaving, divergence is zero.



positive divergence



zero divergence for closed loops

Figure 1: Divergence shows source/sink behaviour.

4 Cartesian Derivation

Vector Field

Let

$$\mathbf{A} = A_x \hat{\mathbf{x}} + A_y \hat{\mathbf{y}} + A_z \hat{\mathbf{z}}.$$

Take a small rectangular box of volume:

$$\Delta V = dx \, dy \, dz.$$

Net Flux

Net outward flux in x direction:

$$\left[\left(A_x + \frac{\partial A_x}{\partial x} dx \right) - A_x \right] dy dz = \frac{\partial A_x}{\partial x} dx dy dz.$$

Similarly:

$$\frac{\partial A_y}{\partial y} dx dy dz$$

and

$$\frac{\partial A_z}{\partial z} dx dy dz.$$

Therefore:

$$\oint_S \mathbf{A} \cdot d\mathbf{S} = \left(\frac{\partial A_x}{\partial x} + \frac{\partial A_y}{\partial y} + \frac{\partial A_z}{\partial z} \right) dx dy dz.$$

Dividing by volume:

$$\nabla \cdot \mathbf{A} = \frac{\partial A_x}{\partial x} + \frac{\partial A_y}{\partial y} + \frac{\partial A_z}{\partial z}$$

5 Divergence of Magnetic Flux Density

Gauss Law for Magnetism

$$\nabla \cdot \mathbf{B} = 0$$

This means magnetic flux density has no source or sink.

Physical Meaning

Magnetic field lines are always closed loops. There is no isolated magnetic monopole. So net magnetic flux through any closed surface is zero:

$$\oint_S \mathbf{B} \cdot d\mathbf{S} = 0$$

6 Divergence of Current Density

Steady Current

For steady current:

$$\nabla \cdot \mathbf{J} = 0$$

This means no charge accumulation occurs in a small volume.

Continuity Equation

The general continuity equation is:

$$\nabla \cdot \mathbf{J} = -\frac{\partial \rho_v}{\partial t}$$

If the current is steady:

$$\frac{\partial \rho_v}{\partial t} = 0.$$

Therefore:

$$\nabla \cdot \mathbf{J} = 0$$

7 Divergence of Curl

Important Identity

For any vector field $\mathbf{A}$:

$$\nabla \cdot (\nabla \times \mathbf{A}) = 0$$

Physical meaning: pure circulation does not create source or sink.

8 Exam Writing Template

For Divergence of B

Write:

1. Fundamental definition of divergence.
2. Cartesian derivation.
3. Apply to $\mathbf{B}$.
4. State $\nabla \cdot \mathbf{B} = 0$.
5. Explain no magnetic monopole and closed field lines.

For Divergence of J

Write:

1. Meaning of current density.
2. Continuity equation.
3. Steady current condition.
4. Final result $\nabla \cdot \mathbf{J} = 0$.
5. Explain current entering equals current leaving.

9 Final Self-Test for This Phase

Instructions

Answer these without looking at the notes. If you cannot answer a point, revise that section again before moving to the next phase.

1. Can I state the main definition of this phase?
2. Can I draw the main diagram neatly without overlap?
3. Can I explain the physical meaning in two or three lines?
4. Can I write the final formula with correct symbols?
5. Can I connect this topic with at least one past-year question pattern?
6. Can I write a semester-style answer with definition, diagram, derivation and final result?

Completion Rule

This phase is complete only when the concept, diagram, formula and answer-writing pattern are all ready.